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ABOUT ME

I am a proactive Software Engineer and a **2nd-year student** at the **National University of Science and Technology POLITEHNICA Bucharest**, within the Faculty of Industrial Engineering and Robotics. My academic background in robotics complements my technical expertise in building complex digital systems, such as my **flagship** project **Garison — a 2D top-down RPG** featuring a custom systems-architecture with GPU-accelerated shaders and high-speed serialization.

Currently, I serve as a **Programming Mentor at Logiscool**. Teaching Python and algorithmic logic has honed my ability to distill technical concepts into clear, accessible knowledge.

WORK EXPERIENCE

 **Logiscool** – Bucharest, Romania

Programming mentor

[01/10/2025 – Current]

- * **Mentorship & Communication:** Simplified complex programming logic (Python, algorithmic thinking) for students of varying skill levels.
- * **Classroom Leadership:** Managed diverse groups of 2 to 20 students, maintaining high engagement levels while balancing different learning speeds and technical backgrounds.
- * **Logic Optimization & Troubleshooting:** Conducted real-time code reviews and live-debugging sessions, identifying and resolving logical errors in student-led projects.
- * **Curriculum Adaptation:** Adapted abstract computer science fundamentals into practical, visual, and engaging projects.

PROJECTS

[01/06/2022 – Current]

Garison - Solo Developer A technically ambitious **2D top-down RPG** developed over four years of iterative refactoring and systems engineering, scaling a monolithic script-base into a modular, high-performance custom engine.

* **Modular Systems Architecture:** Engineered a robust Entity Finite State Machine (FSM) to manage complex AI decision-making and interaction logic, ensuring 100% decoupling from physics and collision modules.* **High-Performance Persistence:** Architected a JSON & Binary buffer serialization engine capable of deep world persistence, allowing for instantaneous saving/loading of complex memory states and DS Map structures.* **GPU-Accelerated Shader Engineering:** Developed a custom GLSL fragment shader pipeline for atmospheric rendering, managing real-time day/night cycles and dynamic lighting effects with zero CPU overhead.* **Lexical Narrative Engine:** Built a custom string parser and dialogue controller featuring real-time tag extraction for dynamic text effects (shake, wave, color) and interactive branching narrative logic.* **Object-Oriented Inventory Logic:** Designed a constructor-based inventory system utilizing modern GMS 2.3+ structs for item stacking, dynamic slot allocation, and scalable item-attribute management.* **State-Driven World Progression:** Implemented a global Quest Controller and state-map system that manages complex cross-room progression and persistent environmental triggers.

Link: <https://github.com/hexa325/Garison2023>

SKILLS

Skills

Advanced AI-Augmented Workflows * **AI-Assisted Development:** Proficient in integrating Large Language Models into the software development lifecycle to accelerate technical planning, debugging, and structural engineering.

* **Multi-Agent Orchestration:** Skilled in managing complex, multi-agent interactions to execute critical code reviews, adversarial problem-solving, and robust system architectural design.

* **Context Engineering:** Expert in technical memory management and strategic context injection (e.g., Markdown-based context documentation) to guide AI tools in producing precise, project-specific code without hallucinations.

Full-Stack Web Development

* **Core Architecture:** Next.js (App Router, Server Components, SSR/SSG), React, TypeScript, JavaScript (ES6+).

* **Backend & Data Integration:** Node.js (via Next.js Server Actions & API Routes), RESTful API Architecture, JSON Data Modeling, Server-Side Data Fetching.

* **Styling & UI Engineering:** Tailwind CSS v4, shadcn/ui, Advanced CSS3 (Animations, Custom Utilities), Semantic HTML5.

* **State & Performance:** React Server/Client State Management (Hooks, Context), Build Optimization, Web Vitals & SEO Configuration.

Systems & Game Programming

* **Languages:** Python (Advanced/Mentoring level), GML (GameMaker Language), C/C++ (Academic/Robotics), C# (Unity).

* **Graphics:** GLSL (Fragment/Vertex Shaders), GPU-accelerated rendering.

* **Architecture:** Finite State Machines (FSM), Object-Oriented Programming (OOP), Data Serialization (JSON/Binary Buffers).

Engineering & Robotics

* **Hardware/Logic:** Industrial Automation, Control Systems, Robotics Systems Architecture.

* **Tools:** MATLAB, CAD (Basic), Logic-based programming.

DevOps & Tools

* **Version Control:** Git, GitHub (Branching, PRs, Actions).

* **Deployment:** Vercel, GitHub Pages.